Smart Storage & Spoilage Prediction

IoT for Sustainable Agriculture

Introduction

Agriculture is the backbone of food security, but one major challenge farmers face is post-harvest losses due to poor storage conditions. Traditional storage methods cannot detect changes in temperature, humidity, or gases that often lead to hidden spoilage.

With growing demand for sustainable solutions, IoT technology offers an opportunity to monitor storage conditions, give real-time updates, and even predict spoilage before it happens.

Our project aims to create a smart, low-cost, and farmer-friendly storage system that reduces waste, improves food safety, and supports sustainable agriculture.

IEEE Xplore: [IoT based Smart Cold Storage System for Efficient Stock Management]

Literature Review: A Gap in Innovation

Many studies have tried to use IoT and sensors for farm storage. These systems mainly help in checking conditions like temperature and humidity. But most of these solutions are made for big storage units, not for small farmers.

Focus on Cold Storage

Common in industries for bulk food storage, but not available for small-scale farmers.

Limited Sensing

Most utilize basic sensors (temperature, humidity, CO_2), providing only environmental monitoring.

No Predictive Power

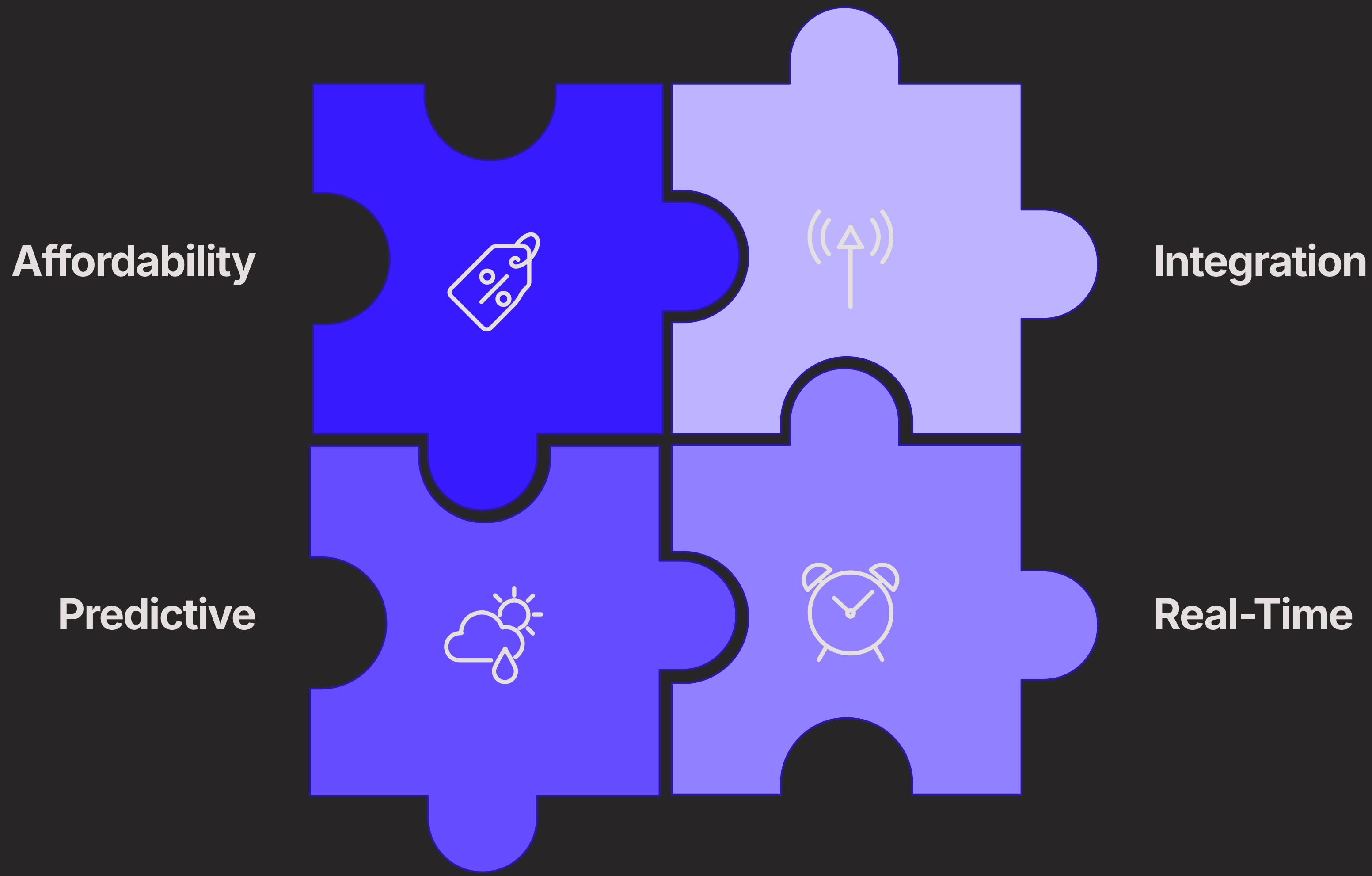
Existing systems can show current storage conditions but cannot predict future spoilage risks.

High Cost & Complexity

Some modern setups exist, but they are costly and complex for local use.

Addressing the Research Gaps

Our project directly tackles the limitations of current solutions, focusing on accessibility, predictive insights and notification warnings to mobile via Blynk app.



We aim to create a system that is not only effective but also practical for small scale farmers, bridging the technology divide in rural communities.

Motivation : Why This Project Matters

We want to reduce food waste, help farmers with easy technology, and create better solutions for a secure food future.

1

Combat Food Waste

Significantly reduce spoilage and financial losses for farmers.

2

Empower Small Farmers

Provide accessible, easy-to-use tools for small scale farmers.

3

Enable Real-Time Decisions

Offer real time warnings via Blynk app.

4

Bridge Tech Divide

Use modern technology to solve rural problems and make sure everyone can benefit from it.

Project Objectives: Measurable Impact

Our objectives are clearly defined, aiming for tangible outcomes that demonstrate the effectiveness and practicality of our solution.

1. **Design & build IoT System**

   Create an IoT-based monitoring system using sensors to track storage conditions like temperature, humidity, and gases.

2. **Real-time Data Monitoring**

   Use Blynk app to provide farmers with live updates of storage conditions anytime, anywhere.

3. **Early Spoilage Prediction**

   Apply threshold-based spoilage prediction on sensor data to detect early signs of spoilage and give timely warnings.

4. **Low-Cost & User-Friendly**

   Develop a system that is affordable, simple to set up, and farmer-friendly.

5. **Simulated Validation**

   Test and demonstrate the system under a simulated storage environment (instead of real warehouses) to prove its effectiveness.

System Architecture: Bringing Smart Storage to Life

Our system combines robust hardware with intuitive software to provide comprehensive monitoring and predictive capabilities.

Hardware Components:

Storage Chamber – Demo box with ventilation

Sensors – DHT22 (Temp & Humidity), MQ-135 (CO_2 & NH_3 gases)

Microcontroller – ESP32 with Wi-Fi

Local Interfaces – OLED display, LED & buzzer for alerts

Software Components:

Programming – Arduino IDE

IoT Platform – Blynk Cloud for dashboards & notifications

Data Processing – ESP32 sends data to cloud, triggers alerts when thresholds are crossed

User Interface – Blynk mobile app with live graphs & instant alerts

System Workflow